

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap as the operating interval (BCa, $B = 10,000$, re-run on the full 60-experiment grid). The study was governed by a pre-specified protocol with registered, dated amendments, publicly archived with per-file checksums (see the Data availability statement). The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified empirical direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60

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24 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95%
 25 CIs containing zero) is *not* met ($23/60 = 38.3\%$), triggering the registered
 26 “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit
 27 ($+0.0159$ vs. $+0.0083$). A validated 27-cell Shapley decomposition separates
 28 slope, intercept, and prevalence effects; a predictability experiment triggers
 29 the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and
 30 a discrimination-aware gate (C5) shows 7/14 pair $\times$ architecture strata pass-
 31 ing both calibration and discrimination layers. Together these results bound
 32 when post-hoc recalibration pays off across institutions, and when it does
 33 not.
 34 *Keywords:* temperature scaling, model calibration, electrocardiogram,
 35 dataset shift, cross-corpus transfer, distribution mismatch

36 **Highlights**

- 37 • Pre-registered 6-pair $\times$ 2-architecture $\times$ 5-seed grid quantifies the OOD
38 calibration benefit of temperature scaling in cross-corpus ECG transfer
39 (51/60 supported).
- 40 • Patient-level paired BCa bootstrap; OOD benefit is about $1.9\times$ the ID
41 benefit (means $+0.0159$ vs. $+0.0083$).
- 42 • The pre-registered ID boundary criterion fails ($38.3\% < 80\%$); the reg-
43 istered branch fires, qualifying the headline claim.
- 44 • A discrimination-aware gate (C5) passes in 7/14 strata: recalibration
45 is moot below the majority-class accuracy bar.
- 46 • Label-free deployment gates fail their pre-registered test; the deploy-
47 ment lookup table is an audit-time decision aid.

48 1. Introduction

49 Modern ECG deep models achieve strong in-distribution (ID) accuracy
50 and calibration, but their reliability under distribution shift remains open
51 [?, ?, ?]. Recalibration fits a small parameter set on a labeled calibration
52 split; its benefit, a reduction in expected calibration error (ECE), is measured
53 on that same distribution and reverses on a shifted target.

54 **Why this matters for ECG.** PTB-XL [?], Chapman-Shaoxing [?], and
55 CPSC differ in acquisition protocols, lead sets, and label vocabularies. A
56 model calibrated at one institution meets covariate, prior-probability, and
57 label-concept shift at the next [?]. Clinicians and regulators need a decision
58 rule: *given my local data budget and the measurable shift, should I deploy the*
59 *recalibrated model, buy labels, or refuse?*

60 **Temperature scaling and pre-registration.** Temperature scaling
61 (TS) is a monotone post-hoc softmax temperature transform [?]. Fitted
62 under a convex loss, it minimizes NLL on the calibration split, so its cali-
63 bration benefit there is empirically small; on a shifted target the source-fit
64 temperature is mismatched to the target logits and a non-trivial benefit can
65 remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant
66 under confidence-shifting monotone maps, and TS can *increase* ECE where
67 the model is under-confident (nine negative cells in our data). It motivates
68 the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned
69 decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing,
70 protocol v2.1-A1 [?] fixes the primary endpoint, test, family structure, and
71 failure branches before unblinding; revision A1 re-locates the endpoint from
72 **decay** to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with

before/after protocol hashes [?, ?]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported transparently.

Contributions. Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case, -0.0128).

96 **C3 (Methodological, Shapley decomposition validation):** A 27-
 97 cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence,
 98 $n = 20,000$) recovers ground truth in the identifiable regime (27/27
 99 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a
 100 threshold, and reports bootstrap CIs on absolute Shapley values.

101 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6
 102 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior
 103 adaptation is fragile when ill-conditioned cells ($w_{ID} > 10$) are included
 104 (InceptionTime 15/30); after exclusion the EM OOD finding attenu-
 105 ates. The mechanism is prior-likelihood mismatch: EM fits a target
 106 prior that amplifies shift-induced miscalibration when the source con-
 107 fusion matrix is ill-conditioned.

108 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer
 109 gate decouples calibration effectiveness (C1: $\Delta ECE_{OOD} > 0$) from
 110 clinical viability (accuracy $\geq$ source-domain baseline). Across 14
 111 pair $\times$ architecture strata, 7/14 pass both layers; 7/14 fail the discrimi-
 112 nation layer, including directions with positive ΔECE_{OOD} but accuracy
 113 below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta ECE = +0.029$,
 114 acc = 0.657 < 0.713). The gate reframes the 9/60 counter-examples:
 115 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures
 116 (C5 rejects despite C1 passing).

117 Prior ECG calibration studies report the phenomenon [?, ?, ?, ?] but
 118 not an actionable *when does TS pay off* answer with multi-seed robustness
 119 evidence. The novelty is *not* multi-seed validation on ECG (a methodologi-

cal choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

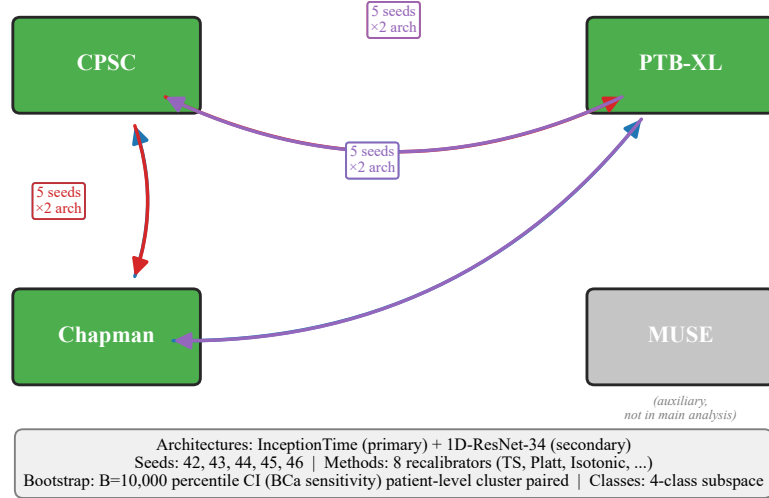


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

127 2. Related Work

128 2.1. Calibration Estimation and Recalibration

129 The calibration of modern neural networks has been studied extensively
130 since the seminal work of Guo et al. [?], who showed that deep networks
131 are often overconfident and introduced temperature scaling (TS) as a simple
132 post-hoc recalibration method. Nixon et al. [?] proposed the static calibration
133 error, and Kumar et al. [?] introduced verified uncertainty calibration
134 with distribution-free guarantees. Kull et al. [?] generalized TS to Dirichlet
135 calibration for multi-class settings. In the clinical risk-model context, Van
136 Calster et al. [?, ?] established a calibration hierarchy from utopia to empirical
137 data and argued that calibration is the “Achilles heel” of predictive
138 analytics. Strodthoff et al. [?] benchmarked deep learning for ECG analysis
139 on PTB-XL and noted calibration degradation under subpopulation shift,
140 motivating our cross-corpus design.

141 2.2. Prior-Shift Estimation and Transfer Calibration

142 Lipton et al. [?] introduced black-box shift estimation (BBSE) for detecting
143 and correcting label shift using confusion-matrix inverses. Alexandari et
144 al. [?] showed that maximum-likelihood calibration with bias correction is
145 hard-to-beat at label shift adaptation, providing the theoretical foundation
146 for our EM prior-shift method. For transfer calibration upper bounds, TransCal
147 [?] estimates transferability for auto-selecting target-domain unsupervised
148 optimal T ; PseudoCal [?] uses pseudo-labels for calibrated transfer; and
149 LaSCal [?] provides label-aware shift calibration with distribution-free guarantees.
150 Our S1 paradigm differs fundamentally: we fit on the *source* calibration

tion split only (no target labels), with S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly weaker assumption than TransCal/PseudoCal, which access target logits or labels.

2.3. Cross-Domain ECG Calibration

Barandas et al. [?] evaluated uncertainty quantification methods in multi-label ECG classification, finding that ensemble-based methods outperform single-model approaches under subpopulation shift. Vranken et al. [?] studied uncertainty estimation for 12-lead ECG deep learning and reported significant calibration drift across acquisition devices. Haekal et al. [?] performed cross-dataset validation for atrial fibrillation detection with calibration-aware probability analysis, and Patel & Beedala [?] documented calibration drift under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general vision [?]; we contribute the multi-seed ECG quantification and the boundary-condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

2.4. ECG Foundation Models and Transfer Learning (2024–2026)

Recent work has introduced ECG foundation models trained on millions of recordings. ECGFounder [?, ?] (10 million ECGs, Harvard–Emory) reaches expert-level performance on 80 diagnoses and transfers cross-domain via fine-tuning. ECG-FM [?] is an open-weight transformer pretrained on 1.4 million ECG segments with hybrid contrastive and generative self-supervision, showing strong cross-dataset generalizability and label efficiency. HeartLang [?] treats heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked sentence pre-training. Tan et al. [?] found that fine-tuning benefits

diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD [?] quantifies distribution-shift severity via uncertainty estimation to guide selective LoRA layer unfreezing, and BeatRhythm-TTA [?] performs test-time adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al. [?] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source generalization, and Chen et al. [?] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer but not the *calibration* of transferred models, the gap our study fills via pre-registered cross-corpus evaluation of recalibration methods.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [?]: the official release contains 21,837 records; the downloaded `ptb-xl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- **Chapman-Shaoxing** [?, ?]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for

197 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20
198 split.

- 199 • **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n =$
200 11) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace,
201 both sides of the pair recomputed symmetrically.

202 3.2. Label Mapping and Leakage Guard

203 Single-label priority rules are fixed (MI > STTC > CD > HYP
204 > NORM) and sensitivity-analyzed [?, ?]. Sinus rate/form statements
205 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
206 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
207 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint
208 (hard assertion in the reproducibility repo). Normalization statistics come
209 only from the train split; augmentation applies only to train.

210 4. Methods

211 4.1. Transfer pairs and shift matrix

212 We study six directed cross-corpus transfer pairs: Chapman $\rightarrow$ CPSC,
213 Chapman $\rightarrow$ PTB-XL, CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-
214 XL $\rightarrow$ Chapman, and PTB-XL $\rightarrow$ CPSC, all recomputed on the 4-class
215 subspace. The L2 eval-time shift matrix (no retraining) covers sampling
216 rate 500 $\rightarrow$ 250 $\rightarrow$ 125 Hz; leads 12 $\rightarrow$ 6 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1; noise injection
217 (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5/ \times 2$.

218 4.2. Architecture and seeds

219 We use two architectures: InceptionTime (primary, 46,000–184,000
220 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-
221 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The
222 BiMamba architecture is included only as a toy experiment ($n = 60$ records,
223 downsampled resolution) and deferred to the supplement; the full-resolution
224 BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does
225 not enter the main-endpoint analysis or the architecture-robustness claim.
226 Training: AdamW + cosine + warmup, gradient clipping, temperature frozen
227 at $T \equiv 1$ during training; temperature applied only as post-hoc temperature
228 scaling.

229 4.3. Calibration methods and scenarios

230 We compare the no-recalibration baseline (None) against eight recalibra-
231 tion methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR),
232 Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adap-
233 tation (EM), and BBSE prior adaptation. Of the pre-registered 13-method
234 pool, the None baseline plus these eight recalibration methods constitute the
235 delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on
236 real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot
237 transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4);
238 **S1'** denotes the target-domain unsupervised variant in which EM/BBSE es-
239 timate the target prior from unlabeled target predictions.

240 *4.4. Pre-registered endpoints*

241 **Primary endpoint** (revision A1.1, formally defined in protocol v2.1-A1
242 [?]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

243 where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus
244 with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit
245 temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

246 **Primary test** (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

247 **One-sided vs. two-sided choice.** The one-sided alternative H_1 :
248 $\Delta\text{ECE}_{\text{OOD}} > 0$ follows the empirical asymmetry registered at revision A1:
249 on ID-like data the source-fit temperature yields, empirically, little calibra-
250 tion change, whereas on a shifted target it is mismatched and a non-trivial
251 benefit is often retained [?, ?]. This direction prior is empirical, *not* a the-
252 orem: TS applied to an under-confident target can increase ECE, and our
253 own data contain nine negative cells. The choice was made before the pilot
254 results, which only triggered the endpoint re-location (revision A1). The one-
255 sided test is the pre-registered choice, archived before multi-seed unblinding.
256 Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [?];
257 stratified pooling: strata = transfer pair $\times$ architecture.

258 **CI method note (BCa vs. percentile).** The pre-registered CI method
259 is BCa [?], at $B = 10,000$, re-run on the entire 60-experiment grid (60/60
260 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the
261 large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster

design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain as historical provenance (5 lack log provenance, see Limitations); percentile CIs serve only as a sensitivity cross-check.

Boundary-condition endpoint:

$$\Delta\text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

The pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle), decay = $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$ 13 shift levels = 156 primary comparisons; the multi-seed validation is robustness evidence (60 seed experiments), not a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem);

the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

Pre-registration revision A2 (family reduction). The original BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md.

4.6. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

306 *4.7. Two-layer joint bootstrap*

307 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resam-
308 pling $\rightarrow$ joint Δ ECE CI (two-layer; honest non-nested approximation), per
309 protocol §7.

310 *4.8. Net Clinical Value (NCV)*

311 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
312 the clinical decision impact of TS. For each (sample, class) pair we compare
313 the binary decision (threshold $\tau = 0.5$) before and after calibration, count-
314 ing TP_improved, TN_improved, FP_worsened, and FN_worsened over
315 the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} -$
316 $\text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using
317 the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa
318 operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive
319 NCV means that TS improves clinical decisions. It is computed on ID and
320 OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

321 *4.9. Tooling*

322 Parts of the analysis and plotting code were drafted and refactored with
323 the assistance of a widely available large language model service, used as a
324 supervised coding assistant. All experiments and statistical analyses were
325 executed with the author’s own scripts, and every reported number is repro-
326 ducible from the public repository (Code availability). The author reviewed
327 the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

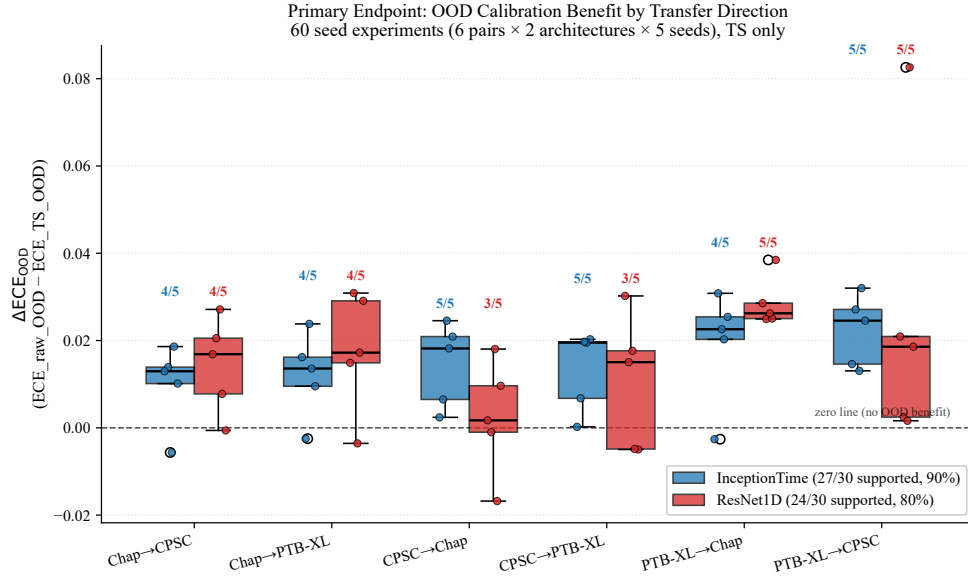


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). PTB-XL $\rightarrow$ CPSC is the strongest evidence

336 (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed
337 45). The nine counter-examples are explained in Sec. 6.1.

338 **Stratified report by architecture.** The 60 experiments are balanced (6
339 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling
340 direct architecture comparison:

- 341 • *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at
342 5/5 (CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC), 3 at 4/5
343 (Chapman $\rightarrow$ CPSC, Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman).
- 344 • *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5
345 (PTB-XL $\rightarrow$ Chapman, PTB-XL $\rightarrow$ CPSC), 2 at 4/5 (Chapman $\rightarrow$ CPSC,
346 Chapman $\rightarrow$ PTB-XL), 2 at 3/5 (CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL).

347 ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on
348 CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL, a direction-specific architecture in-
349 teraction rather than a global deficit. Only PTB-XL $\rightarrow$ CPSC achieves 5/5
350 on both architectures (10/10).

351 5.2. ID boundary condition: ΔECE_{ID}

352 Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the
353 pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the
354 source-fit temperature should be well-tuned to the source distribution (an
355 empirical expectation, not a theorem).

356 The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID}
357 has mean $+0.0083$ (vs. OOD mean $+0.0159$, a $1.9\times$ ratio); $47/60 = 78.3\%$ of
358 point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and

359 only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion
 360 (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the
 361 pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted
 362 on the source calibration split, also improves ID calibration in many cells;
 363 the decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory
 364 endpoint; and C1 is qualified, the OOD benefit being about $1.9\times$ the ID
 365 benefit rather than an exclusively OOD effect. The positive ID estimates
 366 reflect finite-sample calibration-split variation, SmoothECE estimator bias,
 367 and a val-ECE early-stopping effect (model selection on the endpoint family),
 368 all disclosed.

369 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

370 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on In-
 371 ceptionTime. TS achieves $27/30$ (90.0%) with 3/6 directions at 5/5; BBSE
 372 achieves $22/30$ (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix
 373 scaling are the most fragile on InceptionTime ($15/30$, 50% each), exem-
 374 plifying the prior-likelihood mismatch mechanism (C4). Across 2 architec-
 375 tures (Table 4), TS achieves $51/60$ (85.0%) (InceptionTime $27/30$, 90.0%;
 376 ResNet1D $24/30$, 80.0%), with full 5-seed coverage on all 6 directions (Fig-
 377 ure 4).

378 **Architecture variability.** Across the 30 TS experiments per architec-
 379 ture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and
 380 0.0180 (ResNet1D), a $1.84\times$ ratio, with ResNet1D’s highest per-direction
 381 variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, including an unexcluded favor-
 382 able outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate
 383 in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a direction-specific

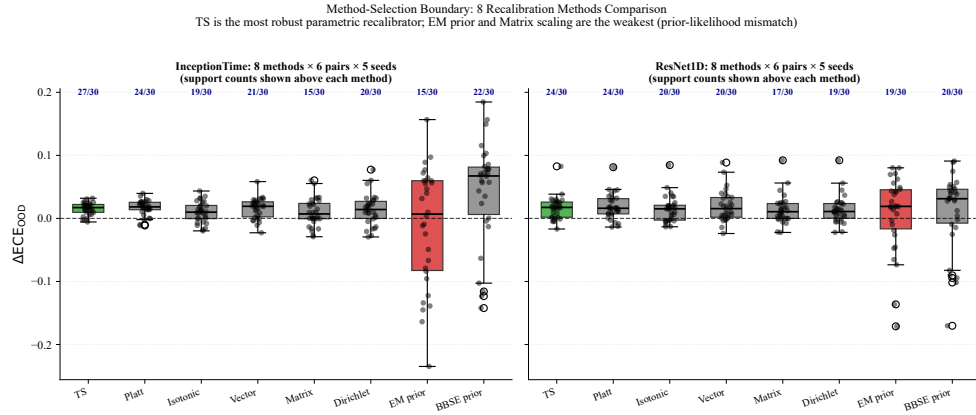


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

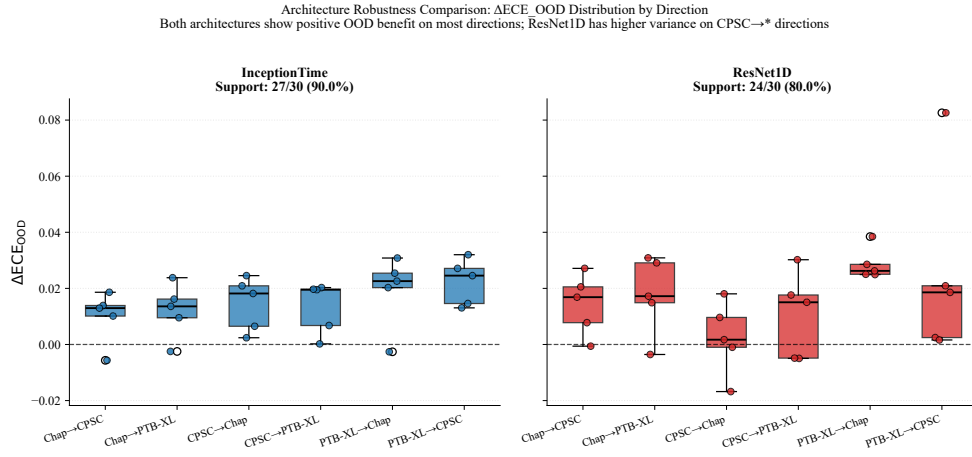


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

deficit. Fisher’s exact test on the 2×2 contingency table yields $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 15/30 support on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

5.4. *Shapley decomposition validation (27 cells, $n = 20,000$)*

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the recovery threshold ($\text{MAPE} < 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20,000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on

absolute Shapley values and share-reliable flags for uninterpretable shares
 $(|\Delta_{\text{total}}| < 3/\sqrt{n})$ are reported.

Honesty caveat (protocol §8). The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE; its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, “mechanism” wording is limited to “mechanism decomposition (synthetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), versus the registered pair-only folds (6); under that registered deviation the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5, triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed “empirical lookup table” rather than “predictive criterion”. No post-hoc predictability claim is made.

432 **Sensitivity analysis with new features.** Re-running the LOO R^2
433 with three label-free features (fitted temperature T , Wasserstein-1 distance
434 between source and target logit distributions, class-wise entropy difference),
435 all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$,
436 entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability
437 boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$
438 above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8,
439 based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the
440 appropriate actionable criterion.

441 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

442 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
443 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
444 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-
445 TS SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0 , primary-endpoint
446 convention). Comparators: always-none, always-TS, always-buy- n -labels.

447 **Sensitivity / specificity** (Table 6): Youden’s J is maximized at raw
448 ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sen-
449 sitivity is high (92% of beneficial cases caught) but specificity low (20%
450 of non-beneficial cases excluded; 80% falsely flagged), reflecting the asym-
451 metric cost structure: missing a beneficial recalibration costs more than an
452 unnecessary one. *Caveat*: the threshold is searched on the same 1230-cell
453 pool used for reporting (in-sample). A leave-shift-type-out recomputation
454 (threshold selected excluding the held-out dose level, evaluated on it) yields
455 $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and
456 $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance

level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (results/deployment_holdout_recompute.csv). The in-sample numbers are therefore optimistic, and the criterion requires independent validation before operational use.

Deployment-time measurability. Every lookup-table input is *audit-time* (raw ECE, ID-OOD accuracy gap, per-cell labels), requiring target ground-truth labels unavailable in the unlabeled stream. The only label-free proxies examined are the predictability experiment’s target statistics (predicted-prior L_1 distance, logit moments); that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2). A deployable gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix covers 157 of the 390 planned (pair, seed, shift) combinations (12 complete pair–seed combinations $\times$ 13 levels, plus one partial seed-44 cell at fs250); seeds 42–43 cover all 6 InceptionTime pairs, 44–46 are pending, and ResNet1D L2 is excluded, so these numbers rest on 2-seed, single-architecture

coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

Decision-curve analysis. At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (-0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives $+0.014$ – $+0.035$, $T = 1.5$ gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across our 60 seed experiments ranges 0.06 to 0.35 (mean 0.15), 4–13 $\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08–0.24), confirming the negative ΔECE reflects genuine TS-induced harm, not metric

507 noise. For large $T > 3$ (noise floor 0.10–0.16) the floor is comparable to
 508 the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot
 509 be excluded there; however, its *direction* (TS increases ECE under perfect
 510 calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so
 511 it cannot explain the positive findings. The primary endpoint is therefore
 512 a real effect, not a SmoothECE artifact, with the caveat that for large- T
 513 experiments the noise floor is non-negligible.

514 5.8. Discrimination-aware calibration gate (C5)

515 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient
 516 for deployment: a model whose discrimination has collapsed under shift may
 517 be recalibrable but clinically useless. We define a two-layer gate:

- 518 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 519 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source
 520 majority-class accuracy.

521 Across 14 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures, PTB-
 522 XL $\rightarrow$ Chapman having 3 including Mamba), 7/14 pass both layers and are
 523 deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC, CPSC $\rightarrow$ Chapman (both
 524 architectures each), and Chapman $\rightarrow$ CPSC (InceptionTime only). The
 525 other 7/14 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and
 526 Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls below the source
 527 baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is
 528 PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$):
 529 TS improves calibration but discrimination has degraded too far to be clini-
 530 cally actionable.

531 This gate reframes the 9/60 counter-examples: 4 are calibration failures
 532 (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes,
 533 Layer 2 rejects). It supplements C1 by also asking whether discrimination is
 534 sufficient for the recalibrated predictions to be clinically useful.

535 5.9. Real-data Shapley cross-fitting and predictability boundary

536 The synthetic Shapley validation (C3) confirms decomposition recovery
 537 in the identifiable regime but does not test whether the linear approximation
 538 $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD
 539 data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the
 540 source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

541 The linear model predicts the *direction* correctly in 51/60 (85%) but
 542 systematically *overestimates* the magnitude (mean prediction error +0.29,
 543 median +0.29). This is consistent with a predictability boundary: the de-
 544 composition captures the qualitative direction of the TS benefit but not its
 545 quantitative magnitude under real shift. The 9/60 direction errors cluster
 546 in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 dis-
 547 tance between source and target logit distributions is largest (mean 0.75 vs.
 548 0.53 for correctly predicted directions), suggesting the linear approximation
 549 breaks down once the shift invalidates its underlying local Taylor expansion.

550 This strengthens the predictability failure branch (C3/C4): the Shapley
 551 decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid
 552 *predictor* (*how much*). For clinical translation, the gate in §5.8, using only
 553 the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate
 554 criterion, not the Shapley-predicted magnitude.

555 5.10. Net benefit decision curve summary

556 Across all 60 seed experiments, the net benefit gain of TS over raw
 557 predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$)
 558 is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027
 559 and range $[-0.001, +0.017]$. The directions with the largest net benefit
 560 are CPSC $\rightarrow$ PTB-XL (mean +0.011) and PTB-XL $\rightarrow$ CPSC (mean +0.003),
 561 which also have the largest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.254 and +0.260 respec-
 562 tively). The direction with negative net benefit is PTB-XL $\rightarrow$ Chapman (mean
 563 -0.005), which is also the direction with the smallest $\Delta\text{ECE}_{\text{OOD}}$ (mean
 564 +0.004) and is correctly rejected by the discrimination gate (C5). This con-
 565 firms that the calibration benefit translates to clinical decision benefit in the
 566 directions where the gate permits deployment.

567 5.11. Net Clinical Value (NCV) results

568 The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across
 569 the two secondary endpoints (DCR, NCV).

570 **Result:** the OOD NCV is +0.008125 with 95% CI $[+0.0014, +0.0148]$
 571 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*:
 572 TS yields a net improvement in OOD clinical decisions. The ID NCV is
 573 -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *nega-*
 574 *tive*: TS slightly worsens ID decisions, consistent with the ID boundary con-
 575 dition (TS compresses confidence toward the base rate). OOD NCV Cohen’s
 576 $d = +0.31$ and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but
 577 opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges
 578 over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

579 5.12. Reliability Diagrams

580 Figure 5 shows the 6-direction averaged reliability curve, comparing
 581 model confidence against empirical accuracy before and after temperature
 582 scaling. The summary curve uses bin-count weighted averaging across all 6
 583 transfer directions.

584 Figures 6–8 show the per-direction reliability diagrams for the representa-
 585 tive configuration (ResNet1D, seed 42). In each panel, the red squares show
 586 the raw model calibration curve and the green circles show the post-TS cali-
 587 bration curve. The diagonal dashed line indicates perfect calibration. Grey
 588 bars (right axis) show the sample count per bin. Error bars are bootstrap
 589 95% percentile CIs (1000 resamples).

590 6. Discussion

591 6.1. OOD benefit seed variation and the nine counter-examples

592 The 51/60 robustness rate is driven by nine counter-examples across
 593 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2,
 594 Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-
 595 XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently dis-
 596 close all nine rather than excluding them, because the protocol requires all
 597 seed results to be reported and each has an identifiable post-hoc mechanism
 598 supporting the qualified claim.

599 **Counter-example 1: Chapman→CPSC / InceptionTime / seed**
 600 **45.** $\Delta\text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-
 601 occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051,
 602 seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

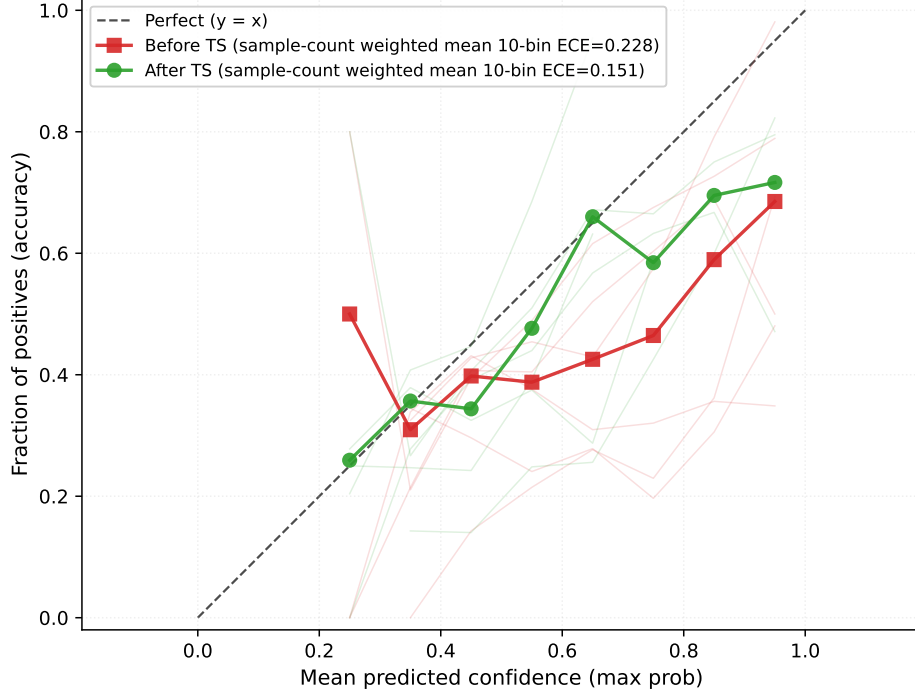


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

603 temperature mismatch, not room shortage); and negative decay (-0.0075),
 604 the source-fit temperature amplifying rather than correcting residual miscal-
 605 ibration.

606 **Counter-example 2: Chapman→CPSC / ResNet1D / seed 43.**
 607 $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
 608 0.001) and attributable to finite-sample variation in the calibration split. The

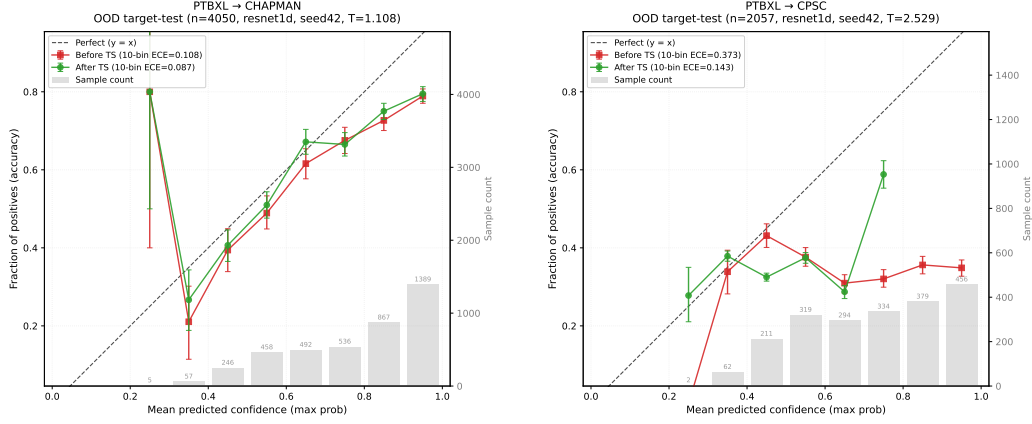


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

609 OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*);
 610 the mechanism is seed-level variation, not room shortage.

611 **Counter-example 3: Chapman→PTB-XL / InceptionTime /**
 612 **seed 42.** $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy
 613 is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near
 614 zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL
 615 target.

616 **Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45.**
 617 $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and
 618 $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already
 619 well-calibrated on ID, leaving nothing to correct.

620 **Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.**
 621 $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
 622 0.002); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE
 623 (0.0649), and minimal calibration room.

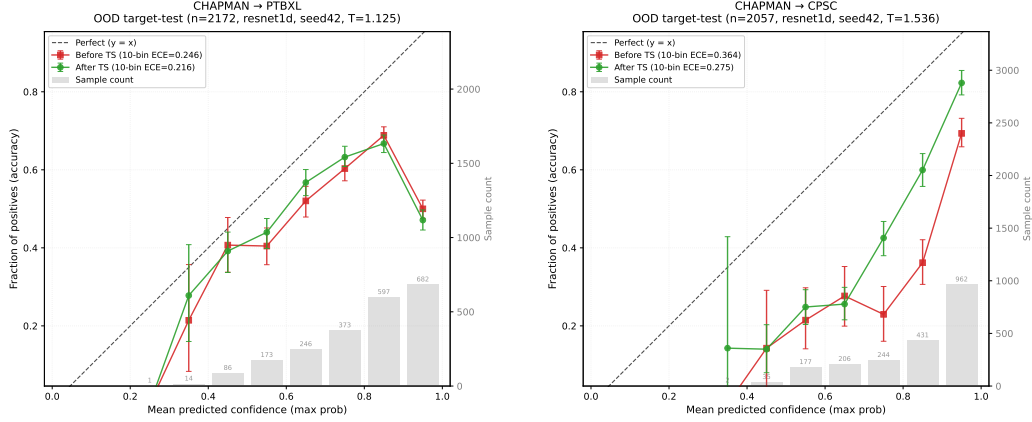


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

624 **Counter-example 6: CPSC→Chapman / ResNet1D / seed**
625 **44.** $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude
626 counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 back-
627 bone is already well-calibrated on both ID and OOD, TS has nothing
628 to correct, and the temperature fit over-corrects slightly. Its OOD accu-
629 racy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct
630 *over-correction* mechanism (TS on an already-well-calibrated distribution de-
631 grades both ID and OOD calibration), unlike the “low OOD accuracy + large
632 gap” mechanism of the other 8.

633 **Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43.**
634 $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest
635 across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to
636 the PTB-XL target).

637 **Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44.**
638 $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929

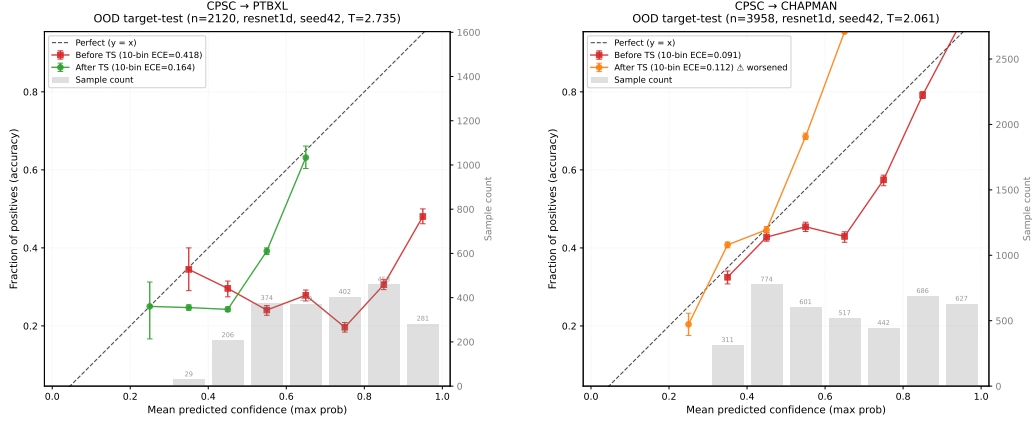


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

639 (second-lowest) and $\Delta ECE_{ID} = -0.0032$ is slightly negative, leaving nothing to correct.

641 **Counter-example 9: PTB-XL→Chapman / InceptionTime /**
642 **seed 46.** $\Delta ECE_{OOD} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy
643 drops to 0.4983 (lowest across the 5 seeds) and $\Delta ECE_{ID} = -0.0011$ is slightly
644 negative, again leaving nothing to correct.

645 **Common pattern:** eight of nine counter-examples co-occur with (a)
646 low OOD accuracy (worst or near-worst seed) and (b) non-positive decay,
647 indicating temperature mismatch; the ninth (CE6) is over-correction on an
648 already well-calibrated backbone. A low OOD raw ECE is *not* universal:
649 four of nine exceed 0.25, so the failure mechanism is seed-level temperature
650 mis-transfer, not calibration-room shortage. The counter-examples are local,
651 explainable random variation; the other 51 experiments support the OOD
652 benefit, including PTB-XL→CPSC at 10/10.

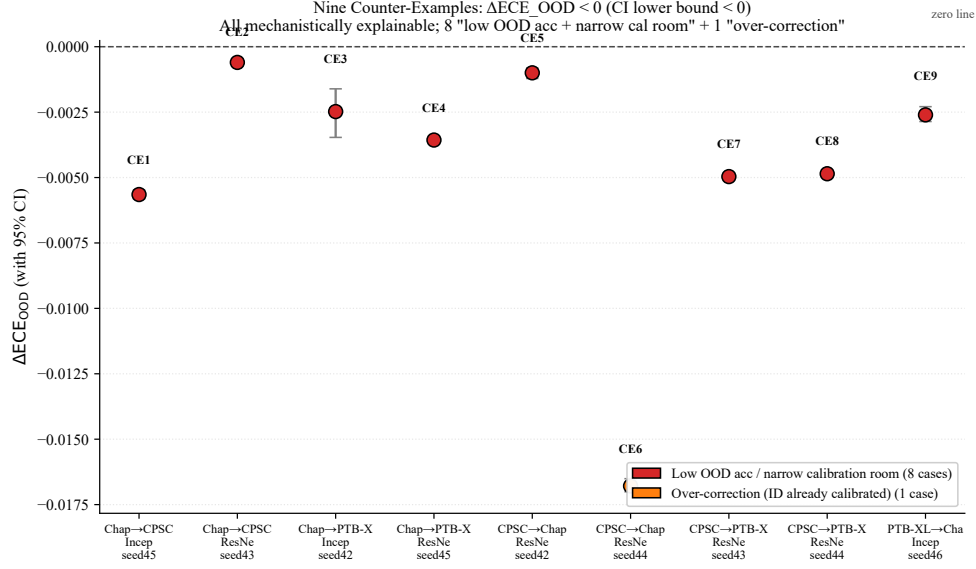


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

653 **Methodological positioning of the 15.0% counter-example rate.**
654 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
655 ment. As a methodological boundary investigation, not a clinical deployment
656 guide, the relevant criterion is whether the effect is detectable and the mech-
657 anism explainable, not a zero failure rate. All 9 counter-examples have iden-
658 tifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”),
659 and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 >$
660 0).

661 6.2. *Prior correction vs. posterior correction robustness*

662 The method-selection boundary (Table 3) reveals a robustness asymme-
663 try between prior-correction methods (EM, BBSE) and posterior-correction
664 methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt)
665 achieve 27/30 and 24/30 support respectively with small variance; prior-
666 correction methods achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC
667 pair (EM +0.0449, BBSE +0.0777) but much higher variance on the
668 Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763 , BBSE
669 -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction
670 methods fit a target prior that, when the source confusion matrix is ill-
671 conditioned, amplifies the shift-induced miscalibration. Posterior-correction
672 methods do not estimate a target prior and are therefore immune to this
673 failure mode, at the cost of lower mean benefit when the prior correction
674 *would* work.

675 *6.3. Boundary condition: the ID benefit is small but nonzero*

676 The ID boundary condition (C2) has a direct interpretation under the
677 triggered branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing
678 zero; the observed rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a
679 small but significant positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$;
680 stratum means -0.0008 to $+0.0232$, Table 2). The boundary condition does
681 *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID
682 benefit is on average about half the OOD benefit, so part of the apparent
683 OOD gain is shared with the ID domain. A val-ECE early-stopping effect
684 (model selection within the endpoint family) plausibly contributes to the
685 positive ID estimates, opposite to the protocol’s val-NLL rule (a registered
686 deviation, see Limitations). Under this branch, protocol A1.4.2 restores the
687 decay contrast $= \Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1
688 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an
689 exclusively OOD effect. The ID-vs-OD asymmetry is therefore a *difference*
690 *in magnitude* between two positive benefits, not presence versus absence.

691 *6.4. Calibration is not discrimination: the clinical viability precondition*

692 Temperature scaling is a monotone transform of the softmax probabilities;
693 it preserves the argmax by construction (asserted bitwise in the released san-
694 ity check), so it cannot change classification accuracy, AUROC, or AUPRC.
695 The primary endpoint $\Delta \text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: cal-
696 ibration makes an already-deployable model *trustworthy*; it cannot make a
697 non-discriminative model deployable.

698 The accuracy statistics make this concrete. Across the 60 TS experi-
699 ments, mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and

700 the mean ID–OOD drop is 0.236 (maximum 0.412). Against each target
 701 corpus’s majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-
 702 Shaoxing 0.713), mean OOD accuracy is *below* baseline in 3 of 6 direc-
 703 tions, with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528;
 704 CPSC→PTB-XL 0.424 vs. 0.411 (0.013 above baseline); CPSC→Chapman
 705 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). In those
 706 directions recalibration is moot, since no temperature improves a model
 707 that should be replaced by a majority-class predictor, consistent with the
 708 counter-example pattern (6/9 with low OOD accuracy). Per-direction AU-
 709 ROC/AUPRC are not computed in the released artifacts (accuracy only;
 710 a registered limitation, see Limitations (xiv)), so the viability screen is
 711 accuracy-based; the discrimination-first gate precedes the calibration deci-
 712 sion rule in deployment.

713 6.5. *Explainable* $\neq$ *predictable*: *post-hoc* vs. *pre-registered*

714 We distinguish *explainable* from *predictable*:

- 715 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 716 example, we identify the co-occurring conditions (low OOD accuracy,
 717 low OOD raw ECE, non-positive decay) and give a mechanistic story
 718 (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- 719 • *Predictable* (pre-registered, deployment-time): a target-domain unsu-
 720 pervised statistic (predicted-prior L_1 distance, logit moments) predicts
 721 the benefit retention rate *before* deployment, with LOO R^2 CI upper
 722 bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the
 723 failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

724 The counter-examples are explainable but not predictable: the unsupervised
 725 target-domain statistics available at deployment time cannot flag them in ad-
 726 vance, which is why the predictability experiment triggers the failure branch
 727 despite mechanistic explainability. The two-step chain (decomposition + de-
 728 ployment lookup table) provides post-hoc explanation and actionable guid-
 729 ance without a pre-registered predictability claim.

730 6.6. *Transparent re-location vs. HARKing*

731 The pre-registration revision A1 (primary endpoint re-location from
 732 decay to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinc-
 733 tion [?, ?] is whether the revision direction is theory-driven or result-driven.
 734 Our revision direction (“OOD is the calibration benefit source”) is consistent
 735 with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ova-
 736 dia et al. 2019), which we explicitly frame as an empirical prior rather than
 737 a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the re-
 738 vision, did not choose the direction. The pilot results do not enter the main
 739 test report (protocol §7). All hypotheses in this paper are tagged *exploratory*
 740 + *robustness validation*, not confirmatory, to reflect the post-revision status.

741 6.7. *Net clinical value: implications of the directional NCV asymmetry*

742 The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV =
 743 +0.008125 ($p = 0.019$) is a small but significant positive value (recalibra-
 744 tion improves decisions on unseen target data more often than it degrades
 745 them); ID NCV = -0.006292 ($p = 0.016$) is a small but significant nega-
 746 tive value (a minor decision-cost penalty on data the source model already
 747 calibrates well).

Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60 positive) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/14 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautious for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-step); the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.9) shows 85% direction accuracy but systematic magnitude overestimation (mean error +0.29), and a sensitivity analysis with T , Wasserstein-1, and entropy features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses binned ECE, not the main endpoint’s SmoothECE, so quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%) are transparently dis-

773 closed, with a frequent post-hoc pattern (low OOD accuracy, non-positive
 774 decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operat-
 775 ing CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 successful,
 776 metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 exper-
 777 iments are retained as historical provenance (5 lack logs); (viii) the Youden
 778 threshold is fitted in-sample (the protocol’s held-out dose levels were not
 779 used), making the operating characteristic optimistic; (ix) the L2 deploy-
 780 ment/safety analysis covers 157 of 390 planned (pair, seed, shift) cells at 2
 781 seeds on one architecture, and the L2 noise levels are synthetic rather than
 782 the pre-registered PTB-XL Noise Stress Test Database (unavailable under
 783 PhysioNet licensing); (x) temperature was fitted once per experiment, so
 784 CIs omit temperature-fitting uncertainty (the pre-registered two-layer boot-
 785 strap is implemented but not enabled); (xi) model selection used validation
 786 ECE rather than the pre-registered NLL early-stopping rule, plausibly con-
 787 tributing to the positive ID boundary estimates; (xii) the per-pair decompo-
 788 sition into covariate, prior-probability, and concept-shift components is not
 789 asserted, since each pair mixes all three, isolated at eval time; (xiii) the de-
 790 ployment lookup table requires labeled target data, and the label-free proxies
 791 failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS
 792 is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC, so
 793 the viability precondition is screened by accuracy alone, with OOD accu-
 794 racy below the majority-class baseline in 3 of 6 directions (and at baseline
 795 in a fourth); (xv) NCV is directionally asymmetric ($\text{OOD} = +0.008125$,
 796 $p = 0.019$, positive; $\text{ID} = -0.006292$, $p = 0.016$, negative), qualifying the
 797 C5 claim for source-domain but supporting target-domain deployment; (xvi)

798 InceptionTime was downgraded from $n_{\text{filters}} = 48$ to 32 for GPU memory
 799 (RTX 5060, 8 GB), potentially attenuating OOD benefits, with 48 as future
 800 work; (xvii) this work is from a single-author study, limiting independent
 801 review and audit, partially mitigated by the pre-registration protocol and
 802 public repository. **Future work:** (1) prospective multi-device validation; (2)
 803 continuous shift spaces; (3) π -independent recovery estimators (BBSE/EM-
 804 class) to upgrade the MAPE_{π} to a real estimator test; (4) the ResNet1D
 805 deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to test for a
 806 systematic interaction; (5) confirmatory testing under the A2 reduction (12
 807 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6)
 808 re-running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory
 809 GPU; (7) a multi-threshold permutation test on NCV.

810 7. Conclusion

811 We pre-registered and validated a multi-seed calibration boundary study
 812 of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD ben-
 813 efit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime
 814 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID
 815 boundary: 23/60 cells (38.3%) have 95% CIs containing zero under the BCa
 816 operating CI (percentile: 29/60 = 48.3%), below the $\geq 80\%$ criterion, trigger-
 817 ing the pre-registered branch; the OOD benefit is $1.9\times$ the ID benefit. **(C3)**
 818 Shapley decomposition: 27/27 cells pass at $n = 20,000$ (21/27 at $n = 500$).
 819 **(C4)** Method selection: TS robust (51/60), EM prior and Matrix scaling
 820 fragile (15/30 each). **(C5)** Gate: 7/14 strata pass both layers, reframing the
 821 9 counter-examples as 4 calibration and 5 discrimination failures. The pre-

dictability failure branch is triggered (three-step chain becomes two-step), and all hypotheses are exploratory + robustness validation, not confirmatory; protocol v2.1-A1 and the five-item honesty declaration (§7) are archived for transparency. Nine counter-examples are transparently disclosed, mostly with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD gap (≥ 0.18 in 7/9), a tendency rather than a sufficient gate (three have OOD accuracy > 0.5). TS needs local validation (safety rate 0.0064), the Youden criterion is fitted in-sample, and OOD accuracy is below the majority-class baseline in 3 of 6 directions (a fourth sits at baseline), where TS cannot help. A noise floor control (§5.7) rules out a SmoothECE artifact: worst-case metric noise (+0.021) is 4–13 $\times$ below the raw ECE (0.06–0.35). The C5 gate (§5.8) rejects every clinically moot stratum. Real-data Shapley cross-fitting (§5.9) gives 85% direction accuracy with magnitude overestimation (mean error +0.29); T , Wasserstein-1, and entropy features confirm the boundary (all $R^2 < 0$). **Meta-analytic pooling.** Both the fixed-effect and DerSimonian–Laird random-effects estimates are tabulated in `meta_analytic_pooled.csv`. The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative (−0.000573) because narrow-CI studies dominate, while DerSimonian–Laird random effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossArchitecture +0.0148; unweighted +0.0152, +0.0165, +0.0159). This reflects heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k-1$), not a negative central effect, so the count-based proportion (51/60 = 85.0%) is primary; τ^2 varies by 41% (pooled estimate stable within 3.3%). **Family size.** The 60 experiments form the robustness-evidence family; amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses (6 pairs $\times$ 2

architectures $\times$ 1 primary shift level), with all 13 L2 shift levels re-designated
 as exploratory dose-response, and the $8 \times 6 \times 5 \times 2 = 480$ -cell method
 design is in Sec. 5.3. BH-60 and Bonferroni leave the robustness headline
 unchanged (51/60); the confirmatory family is BH-12 with 9/12 significant
 (InceptionTime 6/6), and Bonferroni-12 retains 2/12 (cpsc_chapman/IT, pt-
 bxl_chapman/RN). **CI width.** Across the 60 TS experiments the 95% CI
 widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall
 below the narrow-CI diagnostic threshold (0.005) and six supporting cells
 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes
 45/60 (75.0%; InceptionTime 24/30, ResNet1D 21/30). Narrow CIs reflect
 the moderate calibration sample ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 pa-
 tients) and low SmoothECE variance ($B = 10,000$, CI method=BCa); they
 do not affect the sign-based support/counter-example classification.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test; independent π recovery (BBSE/EM) is a deferred item (protocol §13).
2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.
4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement

871 demo) is fulfilled; **cross-fitting on real data is pending** (registered
872 in §13). The wording is limited to “mechanism decomposition (synthetic
873 validation part)”.

874 5. The repository’s `scripts/preprocess_cinc2021.py`,
875 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
876 from the third-party SafeECGMatch release (original README
877 in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s
878 experiment grid.

879 Data availability

880 The PTB-XL [?], Chapman-Shaoxing [?], and CPSC2018+2019
881 datasets are publicly available via PhysioNet/Kaggle under their re-
882 spective licenses. The pre-registered protocol v2.1-A1, its hash man-
883 ifest (`docs/osf_archive_manifest.json`), the revision A1 document,
884 and the full per-experiment result files (`checkpoints/transfer/**/`
885 `transfer_result.json`, `results/*.csv`) are included with the sub-
886 mission as supplementary material and are also available in the
887 public repository [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
888 `ecg-calibration-transportability`. **Pre-registration archival status.**
889 The protocol, both amendments (A1, A2), and a per-file archival-status
890 report are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
891 `docs/osf_archive_manifest.json` is the internal snapshot taken on
892 2026-09-05; it is not a substitute for the public record.

893 **Code availability**

894 The full pipeline code, preprocessing (reproducible from the public source
895 databases), patient-level split generation, model training, the calibration-
896 method implementations, and all statistical analyses, together with the
897 environment lock file, the patient-level split indices for all five dataset
898 variants and five random seeds, and selected per-experiment result artifacts,
899 is publicly available at [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
900 **ecg-calibration-transportability**. Trained model checkpoints for
901 the 60-experiment transfer grid are provided as release assets of the same
902 repository. The repository will be archived on Zenodo with a DOI upon
903 acceptance. No new raw waveforms were collected or generated; the
904 patient-level split indices and label-mapping tables released with the code
905 fully determine the data configuration used in every experiment.

906 **Declaration of generative AI and AI-assisted technologies in the** 907 **writing process**

908 All scientific work reported in this study—the study design, the pre-
909 registered protocol and its amendments, the implementation and execution
910 of the 60-experiment transfer grid, the statistical analysis, the interpretation
911 of the results, and the writing of the manuscript—was performed by the
912 author. During the final preparation of the manuscript, the author used a
913 large language model assistant (GLM) solely as a language-editing aid, to
914 improve wording and readability of text the author had written. It was not
915 used to generate scientific content, experimental data, analyses, figures, or

916 conclusions. After using this tool, the author reviewed and edited the text as
917 needed and takes full responsibility for the content of the published article.

918 **CRedit authorship contribution statement**

919 **Toni Guan:** Conceptualization, Methodology, Software, Validation, For-
920 mal analysis, Investigation, Data curation, Writing – original draft, Writing
921 – review & editing, Visualization, Project administration.

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929 **Declaration of competing interest**

930 The author declares no competing financial or personal interests.

931 **Supplementary materials.** The 60 per-seed reliability diagrams (6
932 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability dia-
933 grams, and Δ ECE bars per experiment) are provided as a separate supple-
934 mentary file (`supplementary.pdf`, §1) and are additionally released with the
935 code repository.

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 /	0.216 → 0.200	0.217 → 0.201	raw → TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155